Embodied Music Pedagogy

Introducing a dynamical embodied perspective to music learning and teaching

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Embodied Music Pedagogy, Instrumental Music Learning, Music Educational Technology, Creativity



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Embodied Music Pedagogy, Inclusive Music Education, ArtsSciences Education

OVERVIEW

- Embodied Music Pedagogy
- Creativity
- Focus on the learner: a kinemusical approach
- Focus on the music teacher

EMBODIED MUSIC PEDAGOGY





EMBODIED MUSIC PEDAGOGY

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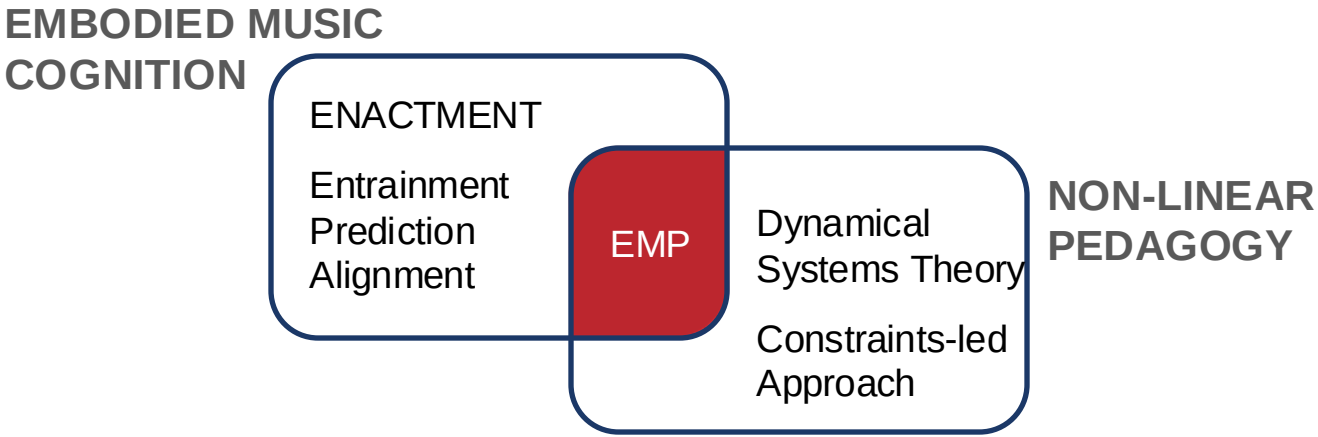
This pedagogy acknowledges the fundamental role
of learners' and teachers' physical and expressive interaction with music and each other, and is developed for lessons in which
communication through/with/in sound and music is the dominant mode.

(Bremmer & Nijs, 2022)

Instrumental & vocal
music learning and teaching

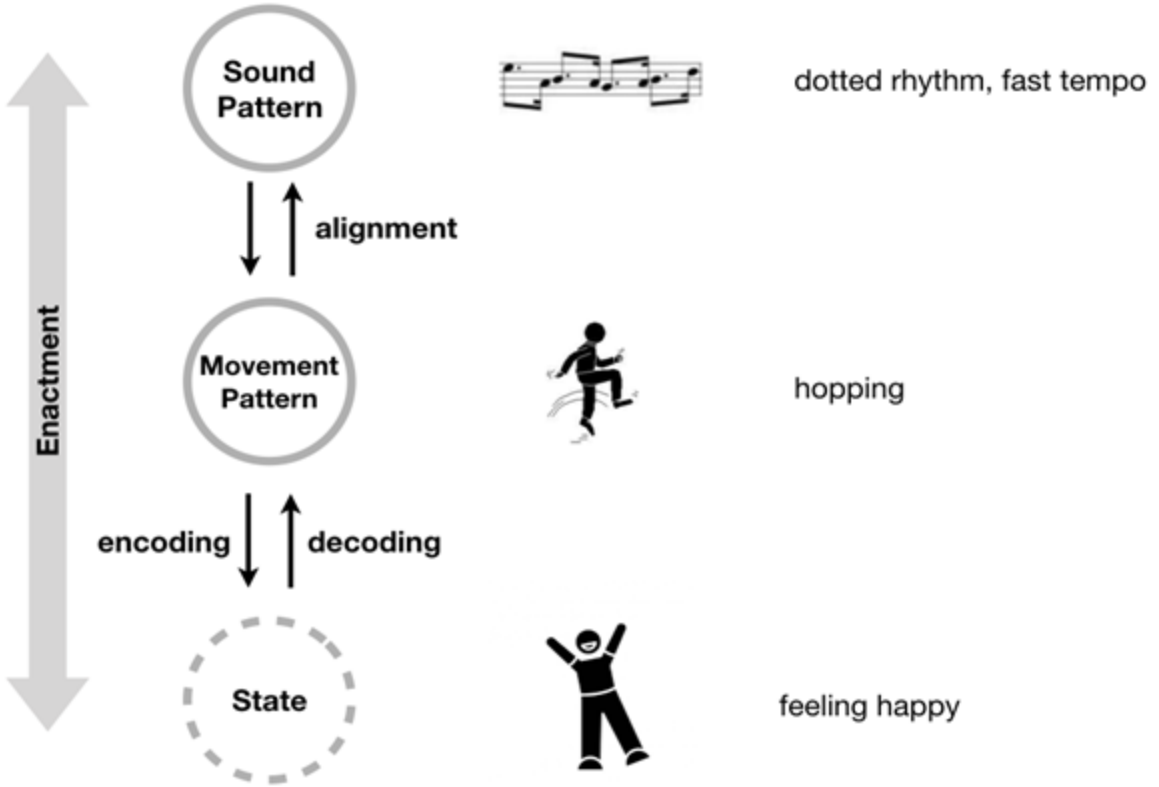
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EMBODIED MUSIC PEDAGOGY





EMBODIED MUSIC COGNITION

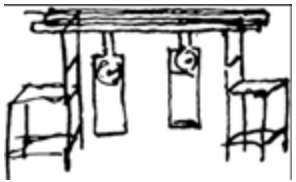

Sound Pattern
dotted rhythm, fast tempo
alignment
Movement Pattern
hopping
Enactment
encoding
decoding
State
feeling happy

Leman, M. (2016)

BASIC MECHANISM:
ENTRAINMENT

Entrainment: ‘the coordination of temporally structured events interaction’ (Clayton et al, 2004)

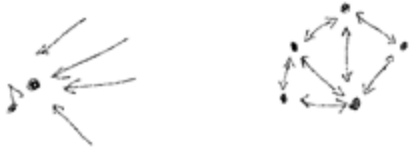
People walking



Heartbeats and clocks

Entrainment in music:

- pulled towards a certain beat in the music
- pulled to the beat of other musicians



Entrainment helps you to develop an underlying time framework

Leman, M. (2016)



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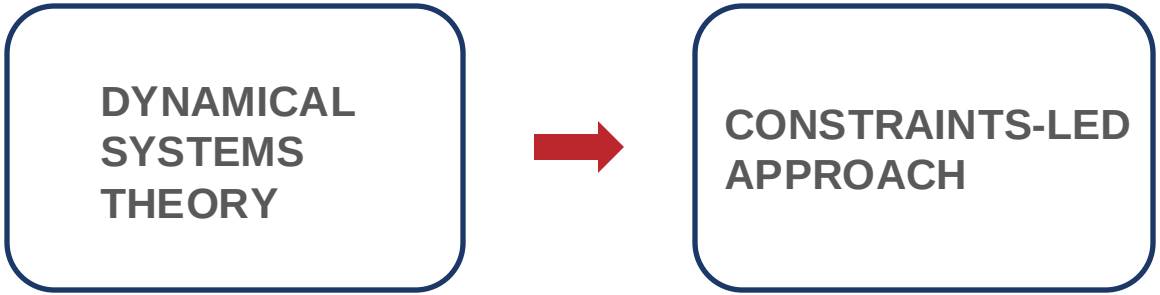
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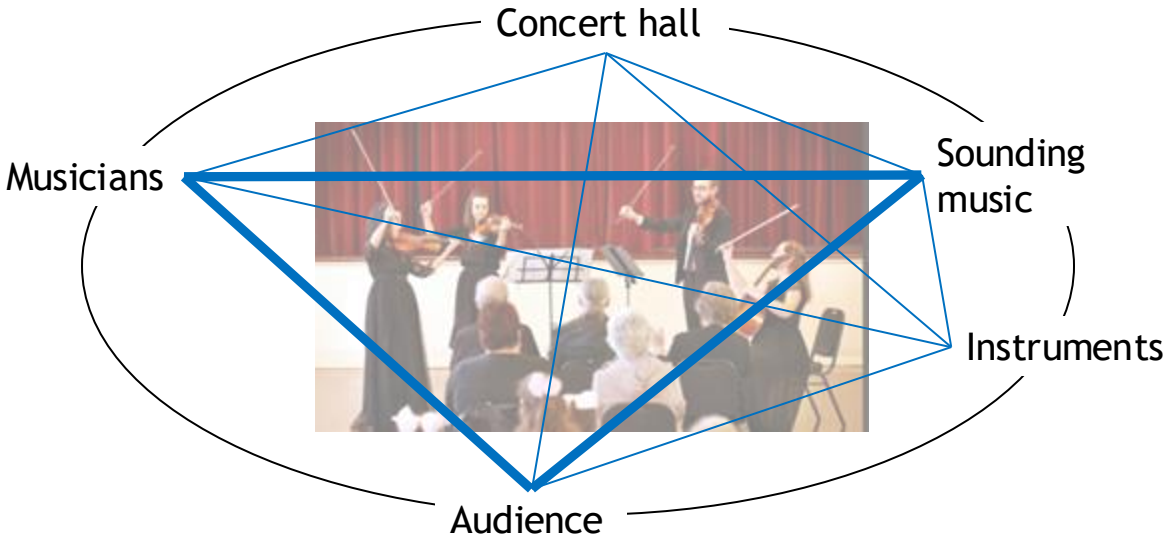
NONLINEAR PEDAGOGY

NON-LINEAR PEDAGOGY



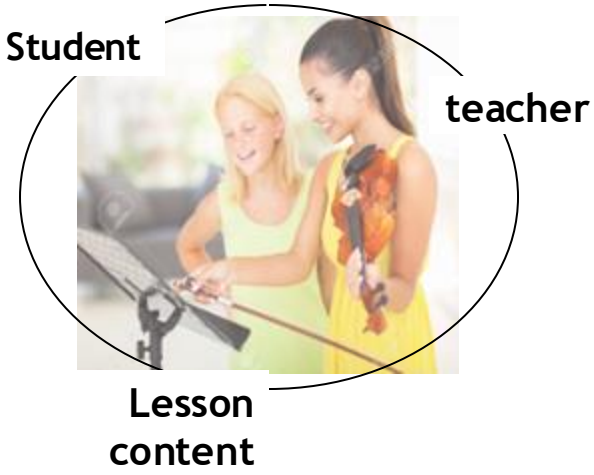
DYNAMICAL SYSTEM THEORY

Systems where the whole is always more than the sum of its parts, based on the dynamic interaction between its components, and between components and the whole (Thelen & Smith, 2006)



Nijs, L., Bremmer, M., Van der Schyff, D. & Schiavio, A. (2023)

DYNAMICAL SYSTEM THEORY & MUSIC EDUACTION



Bremmer, M. & Nijs, L. (2022, 2024).

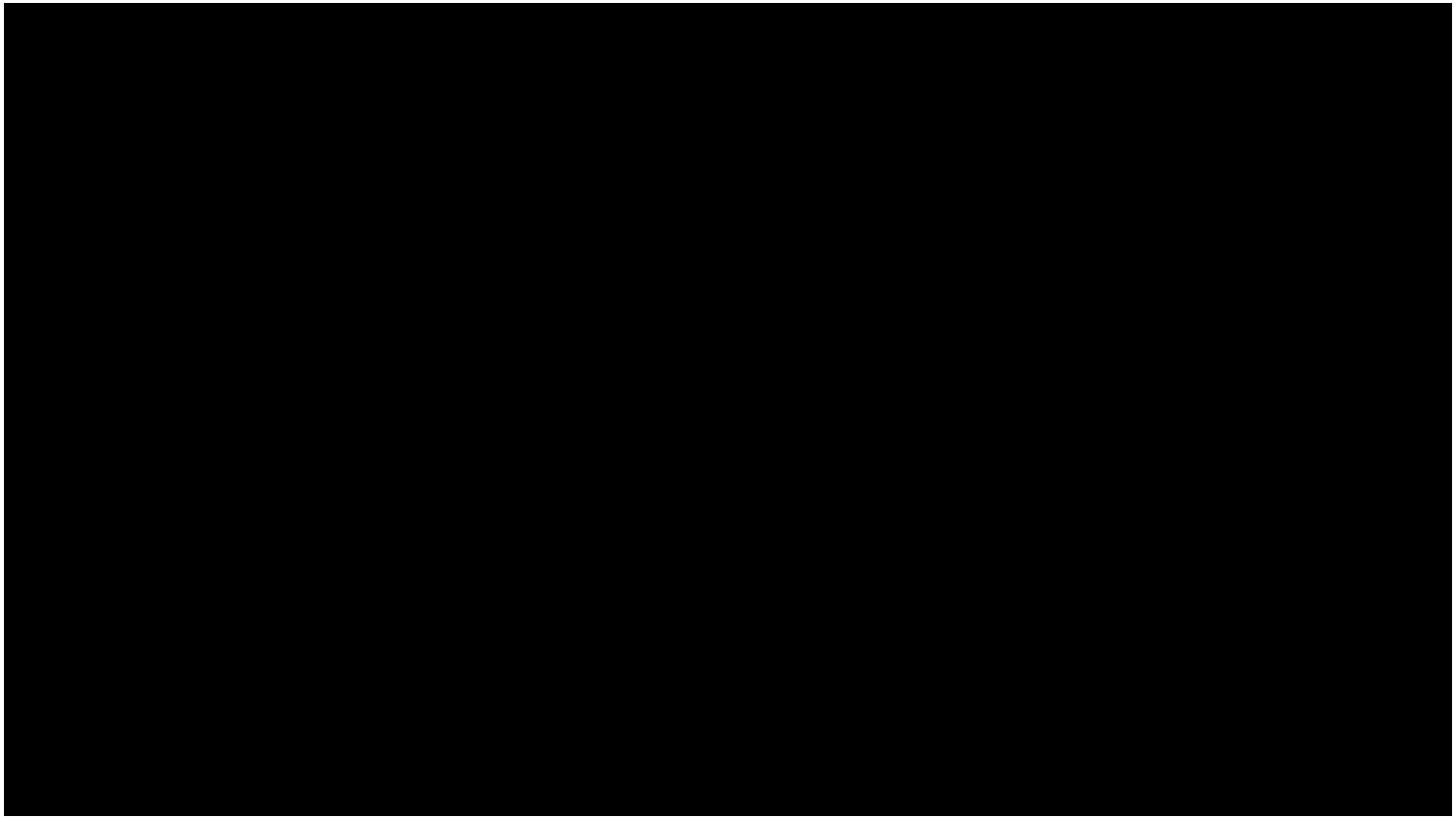
CONSTRAINTS-LED APPROACH

Constraint-led pedagogy revolves around the constraints placed on an individual, shaping or limiting certain their behaviour (Davids, 2012)

Newell (2003) distinguishes between 3 constraints:

- Individual: structural (constant) & functional (change)
- Task: specified by the task to be performed
- Environmental: outside the individual body (incl. socio-cultural)

CONSTRAINTS-LED APPROACH



TO SUMMARIZE

Embodied Music Pedagogy

- addresses students' sensorimotor, affective, creative and cognitive resources
- seeks to deepen the prior musical knowledge through movement-based music activities, and stimulates joint musical sense-making
- invites and challenges to physically experience and explore new musical worlds
- embraces a constraints-led approach


CREATIVITY

CREATIVITY

many different viewpoints with a similar underlying logic:

starts with a novel idea “in the head”

that

“instructs the (mechanical) body to impose the novel form on the material”

creating is not constitutive to the creativity process itself

(Withagen and van der Kamp, 2018)

CREATIVITY

(1)grounds in the unfolding of the action

(2)is relational and

(3)extends into the (social) world.

(Glăveanu, 2012)

CREATIVITY
Instrumental/Vocal Music Education



creativity results from being too focused on children's heads rather than on their bodies, leading to educational approaches that seek to form university professors…



instrumental music education too often focuses on training professional musicians, leading to approaches that emphasise score analysis and instrumental technique in function of adequately reproducing the "master's model of the music" …

CREATIVITY
Rationale

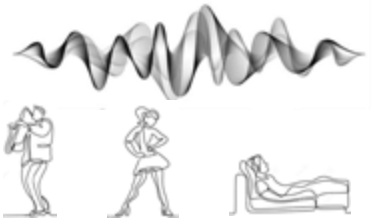
- Music is an affordance-laden structure
- Creativity is about navigating and responding to a rich but complex landscape of musical affordances
- This can be scaffolded by different constraints
- The openness, specification, selection and flexible enaction of competing affordances requires (1) being in a metastable zone, and (2) cognitive flexibility
- An embodied process of musical sense-making (enactment), and its basic mechanisms (entrainment, prediction, alignment) support the flexible navigation of the musical affordance landscape

Nijs, L., Grinspun, N. & Fortuna, S. (2024).

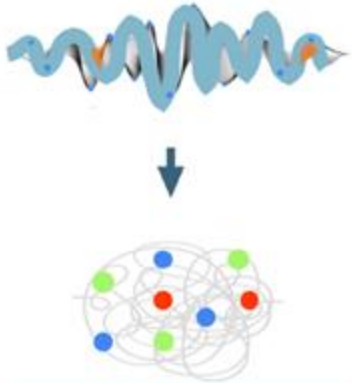
AFFORDANCES

DEFINITION

"Those aspects of music that define what we can do with it" (Krueger, 2014)



STRUCTURE



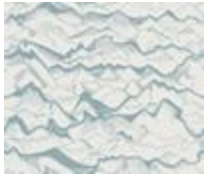
CREATIVITY

"process of perceiving, exploiting and generating novel affordances" (Krueger, 2014)

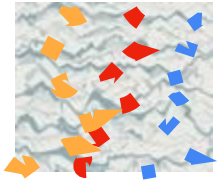


Create conditions explore and experiment in a safe space

NAVIGATING



Music = environment (Reybrouck, 2015)
= virtual space (Nussbaum, 2007)



Navigating = getting optimal grip on the multitude of affordances
(Bruineberg & Rietveld, 2014)

EXPLORE

Change way of listening

↓

Discover hidden affordances

SKILLS

Skilled intentionality

Landscape vs. Field

Affordances vs. Solicitations

CONCERNS & EXPERIENCES

Interest, prefs, needs

Background

METASTABILITY & COGNITIVE FLEXIBILITY

Creativity > creative navigation of the musical environment > openness to different options



Metastability = the ability to possess different co-existent options

- ➛ Context-sensitive selective openness
- ➛ Switch between different responses

Cognitive flexibility

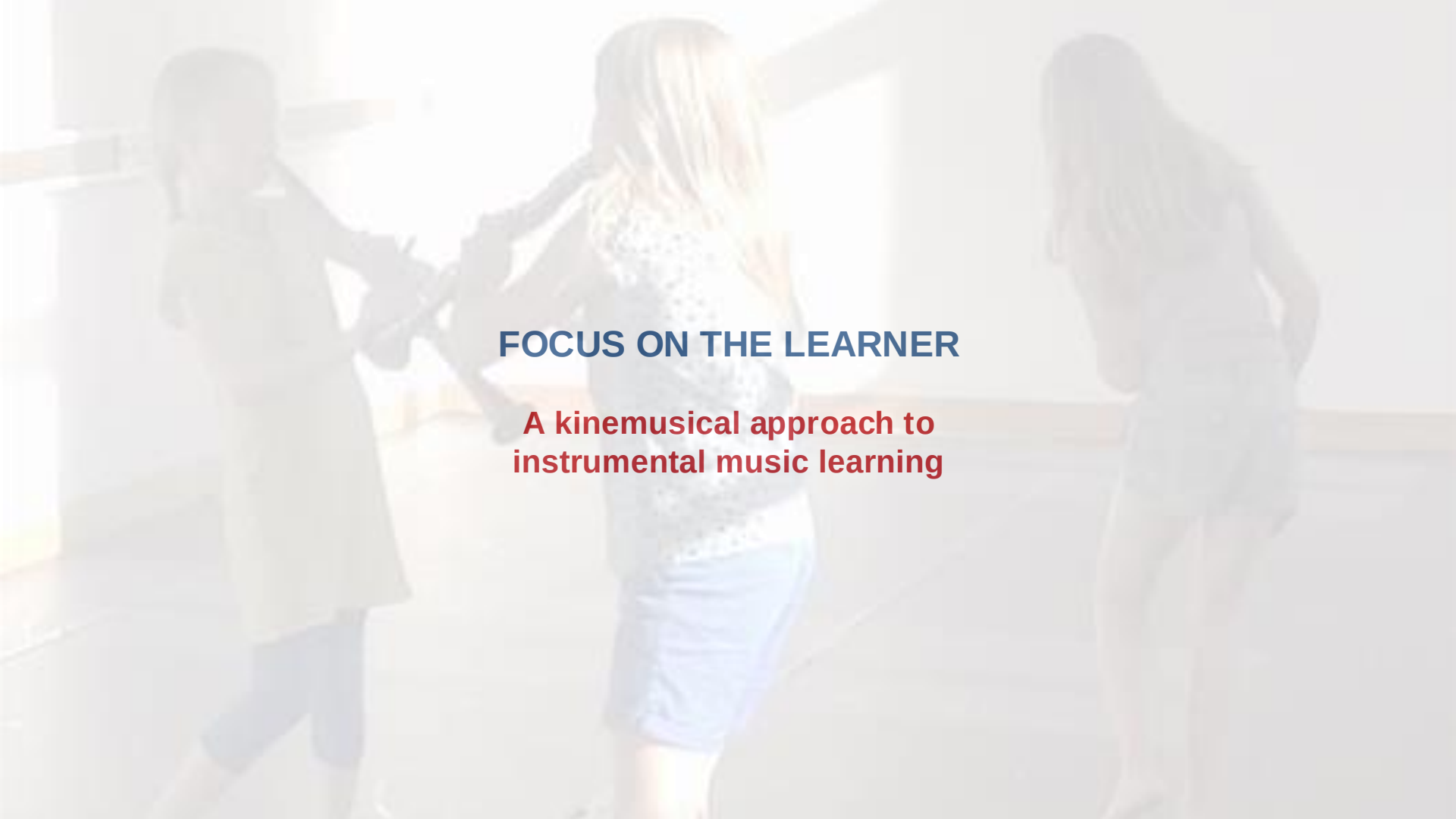
MOVEMENT & CREATIVITY



Create conditions to explore and experiment

Moving to music = creating a metastable zone
= navigating the affordance landscape

Affordances that students perceive, are influenced by teacher's constraints.



FOCUS ON THE LEARNER

A kinemusical approach to instrumental music learning

The students's bodily engagement

Kinemusical approach to instrumental music learning & teaching

reconsider the role of the body and body movement in learning how to play an instrument



Nijs, L. (2017), Nijs, L., M. Lesaffre & M. Leman (2013)

Kinemusical Approach

Developing the inner musician



Involving the instrument



Bottom up



From simple to complex



Exercising the performing body



Kinemusical Approach










Kinemusical Approach
Design Prompts

Musical Activities
- Improvisation
- Performance
- Composition

Spaces
- External
- Personal
- Inner

Spatial Choreography
- Free
- Facing each other
- Circle
- In a row

Social Choreography
- Individual
- One-to-one
- In small groups
- Everyone together

Relationship music-movement
- Exploratory
- Complementary
- Aligning
- Preparatory
- Independent

Social Interaction
- No or little social interaction
- Unidirectional social interaction
- Bidirectional social interaction

Kinemusical Approach



Nijs, L. (2024). *Bewegend lerend musiceren. Een kinemuzikale benadering van het instrumentaal muziekonderwijs*. Garant

Creativity through a Kinemusical Approach

Affordance Navigation

- Body
- Instrument
- Music

Metastability

- Stability
- Flexibility
- Adaptability

Cognitive flexibility

- Shifting between different perspectives
- Choosing between different actions



FOCUS ON THE TEACHER

The teacher's bodily engagement

Musical creativity *includes* performing music (Bremmer & Verneert, 2023)

Music students learn to creatively interpret music

General music education: teacher's bodily involvement seems to support creative music processes (Hendriks et al. 2023; Kupers & Van Dijk, 2020; Chatelain et al., 2024)

Can instrumental and vocal music teachers also enhance students' creativity and expressiveness through their bodily engagement?



The teacher's bodily engagement

Emphasis score analysis and technique (Nijs, 2019)

Reason?: little research on how instrumental and vocal music teachers can stimulate creativity with through bodily engagement

CLA: introduction constraints through teacher's bodily involvement e.g., through pedagogical gestures and pedagogical touch

We will hypothesise how these bodily strategies might scaffold creativity in performance

PEDAGOGICAL GESTURES

(Intuitively) accompany speech or music making to support musical learning process
(Fatone et al., 2011; Nafisi, 2013; Kochman et al., 2014; Simones et al., 2015)

Make the invisible *visible* and provide *extra* information, next to musical and verbal information (Fatone et al., 2011)

Do not necessarily need to be imitated (Simones et al., 2017)

PEDAGOGICAL GESTURES

Technical information: e.g., gestures can visualize a tongue position (Nafasi, 2013a) or demonstrate the resonance space (Kochman et al., 2014)

Musical information: e.g., gestures can visualize the beat of the music (Simones et al., 2015) or the pitch (Pearson, 2016)

AND: **Expressive information:**

- change in loudness and timbre,
- qualities of attack and release (Pearson, 2016)
- articulation, phrasing (Nafisi, 2013b)



Which two types of gestures did you observe?



Lesson with Ronald Rickamp and Monica Monteiro

CREATIVITY & GESTURES

Teachers’ expressive gestures (constraints) can direct learner's attention to less familiar expressive aspects in the music (affordances)

The teachers’ gestures, thus, might *change* the way students listen to music →

offering them opportunities to recontextualize or rework the sound itself, possibly leading to novel interpretations of music (Nijs et al., 2024)

Therefore, we hypothesise that teachers’ expressive gestures might stimulate creativity



TOUCH

Haptic feedback, tactile modeling, molding: similar teaching strategies of manipulating the learner's body (Bremmer, 2015; Metz, 1989)

Touch might be an entry point to facilitate the student’s sense of bodily flexibility and freedom

! Teacher must understand the ethical limits of physical contact as a pedagogical approach (McHugh-Grifa, 2011) !



TOUCH



Teacher moves the student, suggesting new movements

By helping the student to feel a greater sense of bodily flexibility/freedom → more possibilities the navigate the affordances of the musical landscape

Hypothesis: More possibilities to play creatively and expressively



Opening up research vista's

- Need for conceptual clarity: "embodied", affordances, constraints
- Still little research into
  - How movement can stimulate musicians' creativity
  - How instrumental and vocal teachers can use their body in teaching for creativity

→ whole new line of research

- Translation into practice

Follow our work…



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REFERENCES

(In order of appearance)

Bremmer, M. & Nijs, L. (2022). Embodied Music Pedagogy. A vision on the dynamic role of the body in music education. In T. Buchborn, S. Schmid, G. Brunner, and Th. De Baets (Eds.), *European perspectives on music education: Music Is What People Do* (pp.). Helbling.

Leman, M. (2016). *The Expressive Moment: How Interaction (with Music) Shapes Human Empowerment*. MIT Press.

Clayton, M., Sager, R., & Will, U. (2004). In time with the music: The concept of entrainment and its significance for ethnomusicology. ESEM CounterPoint, 1, 1–45.

Thelen, E., and Smith, L. B. (2006). Dynamic systems theories. In W. Damon and R. M. Lerner (Eds.), *Handbook of Child Psychology: Vol. 1. Theoretical Models of Human Development*, 6th ed. (pp. 258–312). Hoboken, NJ: Wiley.

Nijs, L., Bremmer, M., Van der Schyff, D. & Schiavio, A. (2023) Embodying dynamical systems in music performance. Music Performance Research, online first. https://doi.org/10.14439/mpr.11.3

Bremmer, M. & Nijs, L. (2024). Embodiment in music education. *Journal de recherche en éducations artistiques, 2*, Special issue edited by P. Terrien & S. Chatelein, 2, 13-26. https://www.jrea.ch/article/view/4717.

Newell, K. M. (2003). Change in motor learning: a coordination and control perspective. Motriz Rio Claro. 9, 1–6.

Withagen, R., & van der Kamp, J. (2018). An ecological approach to creativity in making. New Ideas in Psychology, 49, 1-6.

Glăveanu, V. P. (2012). What can be done with an egg? Creativity, material objects, and the theory of affordances. T*he Journal of Creative Behavior,* 46(3), 192-208.

Nijs, L., Grinspun, N. & Fortuna, S. (2024). Creativity through movement: navigating the musical affordance space. *Creativity Research Journal*, 1–25. https://doi.org/10.1080/10400419.2023.2299159

REFERENCES

(In order of appearance)

Krueger, J. (2014). Affordances and the musically extended mind. *Frontiers in Psychology, 4*, 1003, 1- 18. https://doi.org/10.3389/fpsyg.2013.01003

Reybrouck, M. (2015). Music as environment: An ecological and biosemiotic approach. *Behavioral Sciences, 5(*1), 1-26. http://doi.org/10.3390/bs5010001

Nussbaum, C. O. (2007). *The musical representation: Meaning, ontology, and emotion*. Cambridge, MA: MIT Press.

Bruineberg, J., & Rietveld, E. (2014). Self-organization, free energy minimization, and optimal grip on a field of affordances. Frontiers in Human Neuroscience, 8, 599. http://doi.org/10.3389/fnhum.2014.00599

Nijs, L. (2017). The merging of musician and musical instrument: incorporation, presence and the levels of embodiment. In M. Lesaffre, P.J. Maes & M. Leman (Eds). *The Routledge Companion to Embodied Music Interaction* (pp. 49 – 57). Routledge.

Nijs, L., M. Lesaffre & M. Leman (2013). The Musical Instrument as a Natural Extension of the Musician. In Castellengo, M. & Genevois, H. (Eds.), *Music and its instruments* (pp. 467–484). Editions Delatour France.

Nijs, L. (2024). *Bewegend lerend musiceren. Een kinemuzikale benadering van het instrumentaal muziekonderwijs*. Garant

REFERENCES

(In order of appearance)

Bremmer, M. & Verneert, F. (2023). De 'creativity turn' in muziekeducatie. In T. de Baets & A. de Vugt (red.), *Beweging in muziekpedagogiek. 10 jaar onderzoek en ontwikkeling* (pp. 13-24). Europrint.

Hendriks, L. H., Steenbeek, H. W., Bisschop Boele, E. H., & van Geert, P. L. C. (2023). Promoting creative autonomy support in school music education: An intervention study targeting interaction. *Frontiers in Education*, *7*, Article 1102011. https://doi.org/10.3389/feduc.2022.1102011

Kupers, E., & van Dijk, M. (2020). Creativity in interaction: The dynamics of teacher-student interactions during a musical composition task. *Thinking Skills and Creativity, 36,* Article 100648. https://doi.org/10.1016/j.tsc.2020.100648

Chatelain, S., Barman, K., Lage-Gómez, C., & Moor, M. (2024). Reframing Creative Teaching in Secondary Music Teacher Education. *Education Sciences* 14(3), Article 324. https://doi.org/10.3390/educsci14030324

Nijs, L. (2019). Moving together while playing music: promoting involvement through student-centred collaborative practices. In S. Gies, & J. H. Sætre (Eds.), *Becoming musicians – student involvement and teacher collaboration in higher music education* (pp. 239-260). The Norwegian Academy of Music.

Fatone, G., Clayton, M., Leante, L., & Rahaim, M. (2011). Imagery, melody, and gesture in cross-cultural perspective. In A. Gritten, & E. King (Eds.), *New perspectives on music and gesture* (pp. 203-220). Ashgate.

Nafisi, J. (2013a). Gesture and body-movement as teaching and learning tools in the classical voice lesson. A survey into current practice. *British Journal of Music Education, 30*(3), 347-367. https://doi.org/10.1017/S0265051712000551

Kochman, K., Moelants, D., & Leman, M. (2014). Gesture as communicative tool in vocal pedagogy. *Journal of Interdisciplinary Music Studies, 6*(2), 233-250.

REFERENCES

(In order of appearance)

Simones, L. L., Rodger, M., & Schroeder, F. (2015). Communicating musical knowledge through gesture: Piano teachers' gestural behaviours across different levels of student proficiency. *Psychology of Music, 43*(5) 723-735. https://doi.org/10.1177/0305735614535830

Simones, L. L. (2017). A framework for studying teachers' hand gestures in instrumental and vocal music contexts. *Musicae Scientiae, 23*(2), 231-249. https://doi.org/10.1177/0305735614535830

Pearson, L. (2016). Gesture in Karnatak music: Pedagogy and musical structure in South India (Unpublished PhD thesis). Durham University, UK. https://etheses.dur.ac.uk/11782/

Nafisi, J. (2013b). *Gesture and body-movement as teaching and learning tools in western classical singing.* (Unpublished doctoral dissertation). Melbourne: Monash University, Australia.

Nijs, L., Grinspun, N. & Fortuna, S. (2024). Creativity through movement: navigating the musical affordance space. *Creativity Research Journal.* https://doi.org/10.1080/10400419.2023.2299159

Bremmer, M. (2015). *What the body know about teaching music* (Unpublished PhD thesis). University of Exeter.)

Metz, E. (1989). Movement as a musical response among preschool children. *Journal of Research in Music Education, 37(*1), 48-60. http://doi.org/10.2307/3344952

McHugh-Grifa, A. (2011). The use of physical touch to facilitate learning in music education. *Visions of Research in Music Education, 18.* Retrieved from http://www-usr.rider.edu/vrme~/